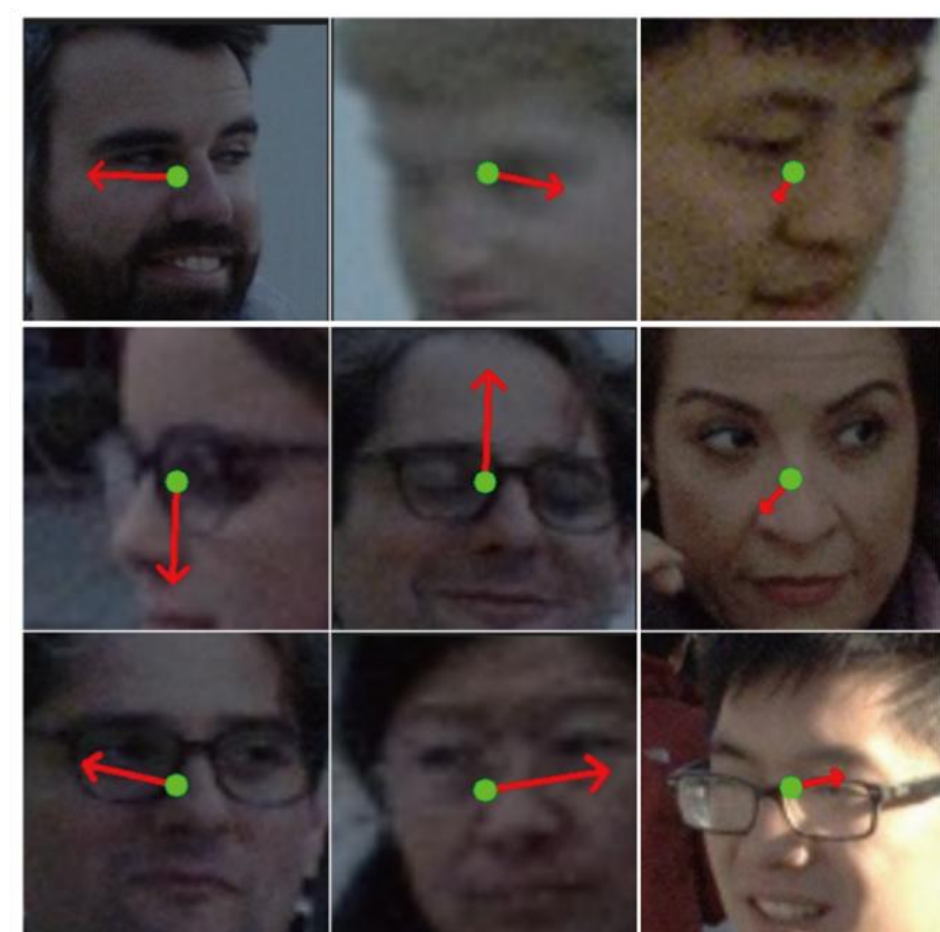




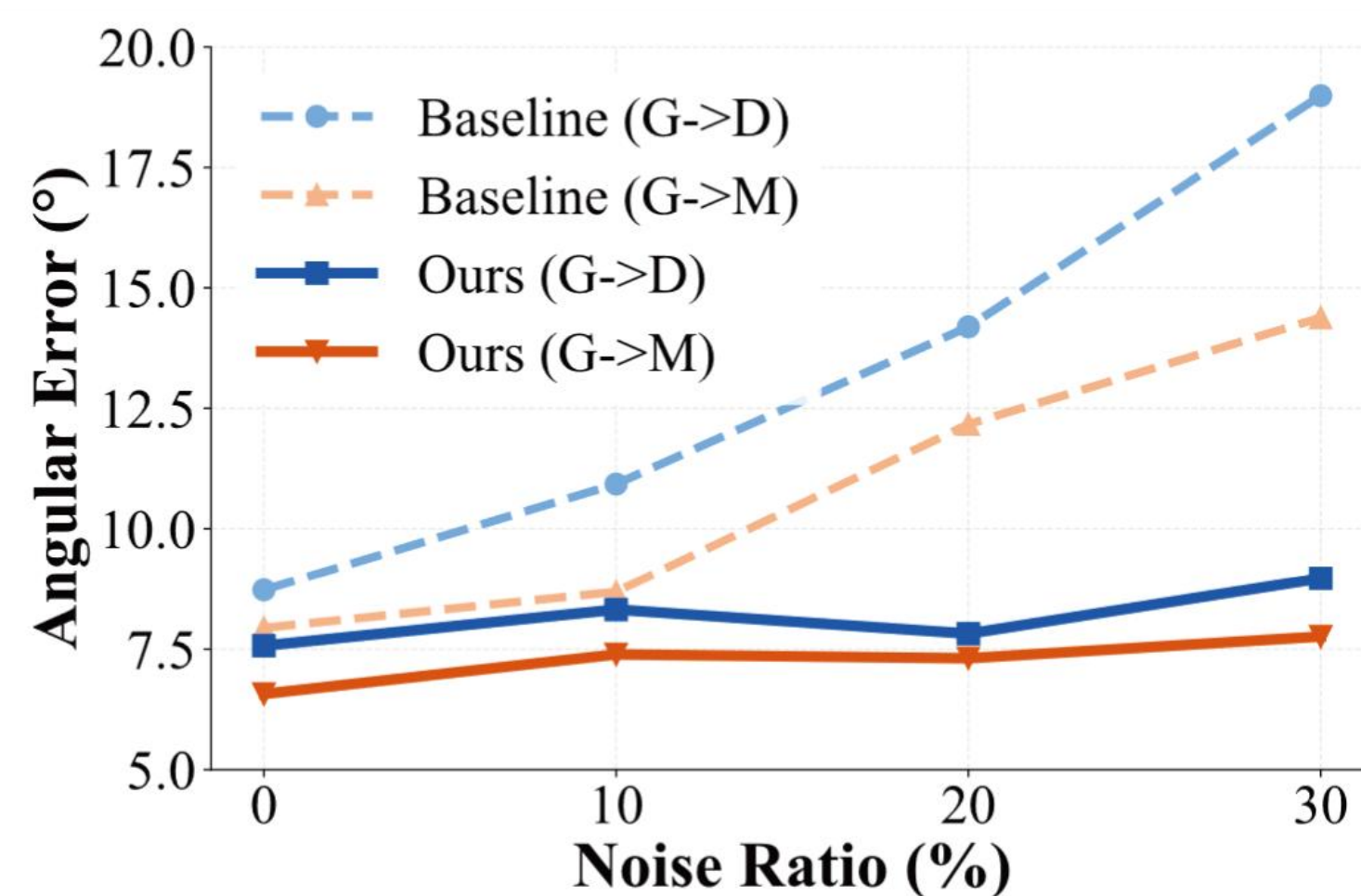
1 Motivation / Problem

- Cross-domain gaze estimation often suffers from poor generalization on unseen domains.
- Large-scale gaze datasets inevitably contain noisy labels due to difficult annotation.
- Existing generalization methods usually ignore label noise.
- Noisy labels make models learn gaze-irrelevant features and harm generalization.

Noisy samples



Noise hurts generalization



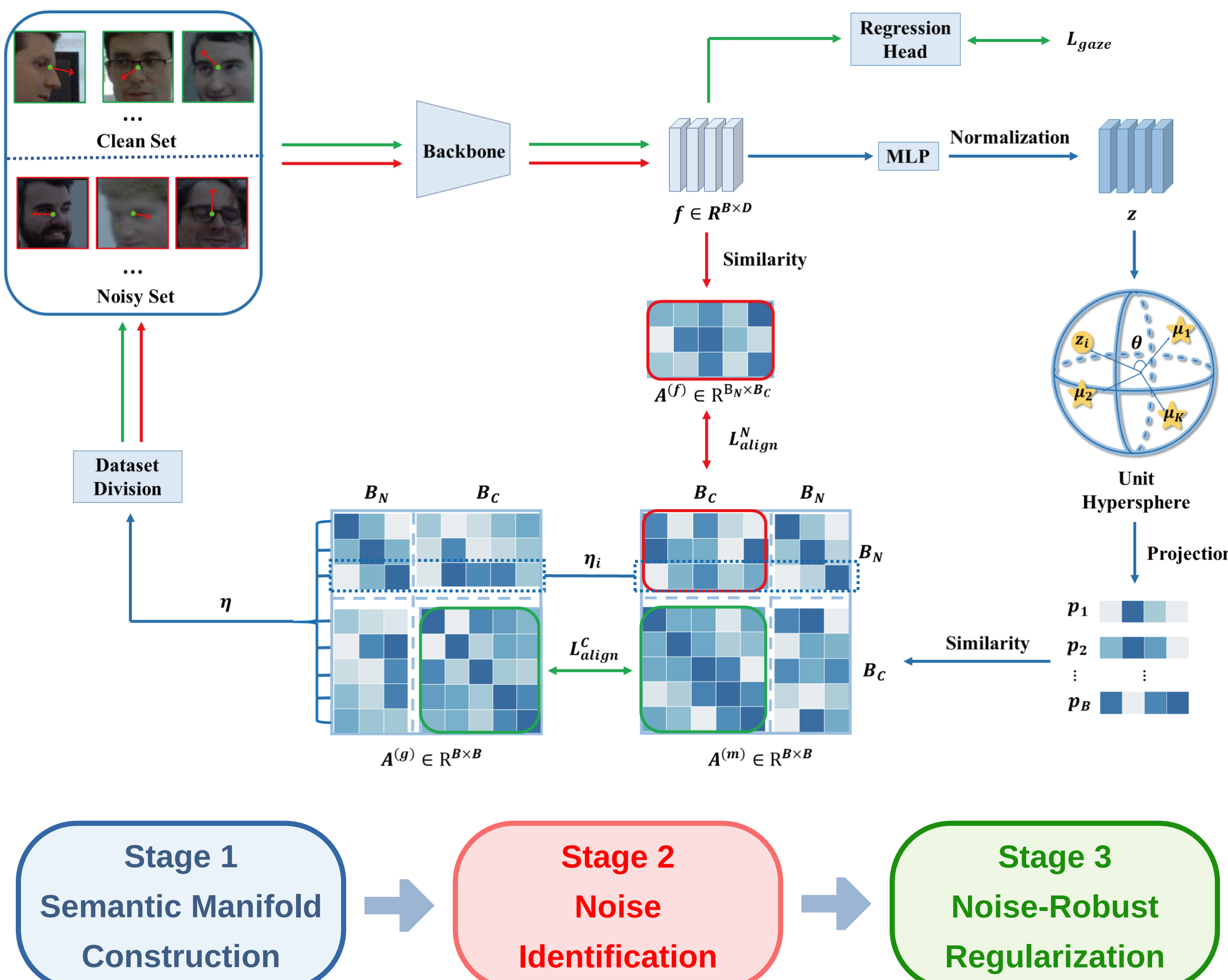
Red arrows: provided labels by Gaze360

On Gaze360→MPIIFaceGaze, error rises from 7.94° to 12.17° under 20% noise, and to 14.38° under 30% noise.

2 Key Contributions

- 1 First study of domain generalization in gaze estimation from the perspective of label noise.
- 2 A semantic manifold aligns continuous gaze label relations with feature relations.
- 3 Affinity-based noise detection and robust regularization improve generalization without sacrificing source accuracy.

3 Method Overview



5 Experiments & Analysis

A) Quantitative Results

Main results (Angular Error °)

Method	$\mathcal{D}_E \rightarrow \mathcal{D}_M$	$\mathcal{D}_E \rightarrow \mathcal{D}_D$	$\mathcal{D}_G \rightarrow \mathcal{D}_M$	$\mathcal{D}_G \rightarrow \mathcal{D}_D$	within $\mathcal{D}_E^*$	within $\mathcal{D}_G^*$
ResNet-18	8.07	8.78	7.94	8.73	4.64	11.14
ResNet-18+SeeTN	6.58 ↓18.5%	7.18 ↓18.2%	6.57 ↓17.2%	7.57 ↓13.3%	4.40 ↓5.1%	10.73 ↓3.7%
ResNet-50	7.64	8.39	7.68	8.65	4.32	10.78
ResNet-50+SeeTN	6.31 ↓17.4%	6.84 ↓18.4%	6.75 ↓12.1%	7.42 ↓14.2%	4.16 ↓3.7%	10.45 ↓3.1%

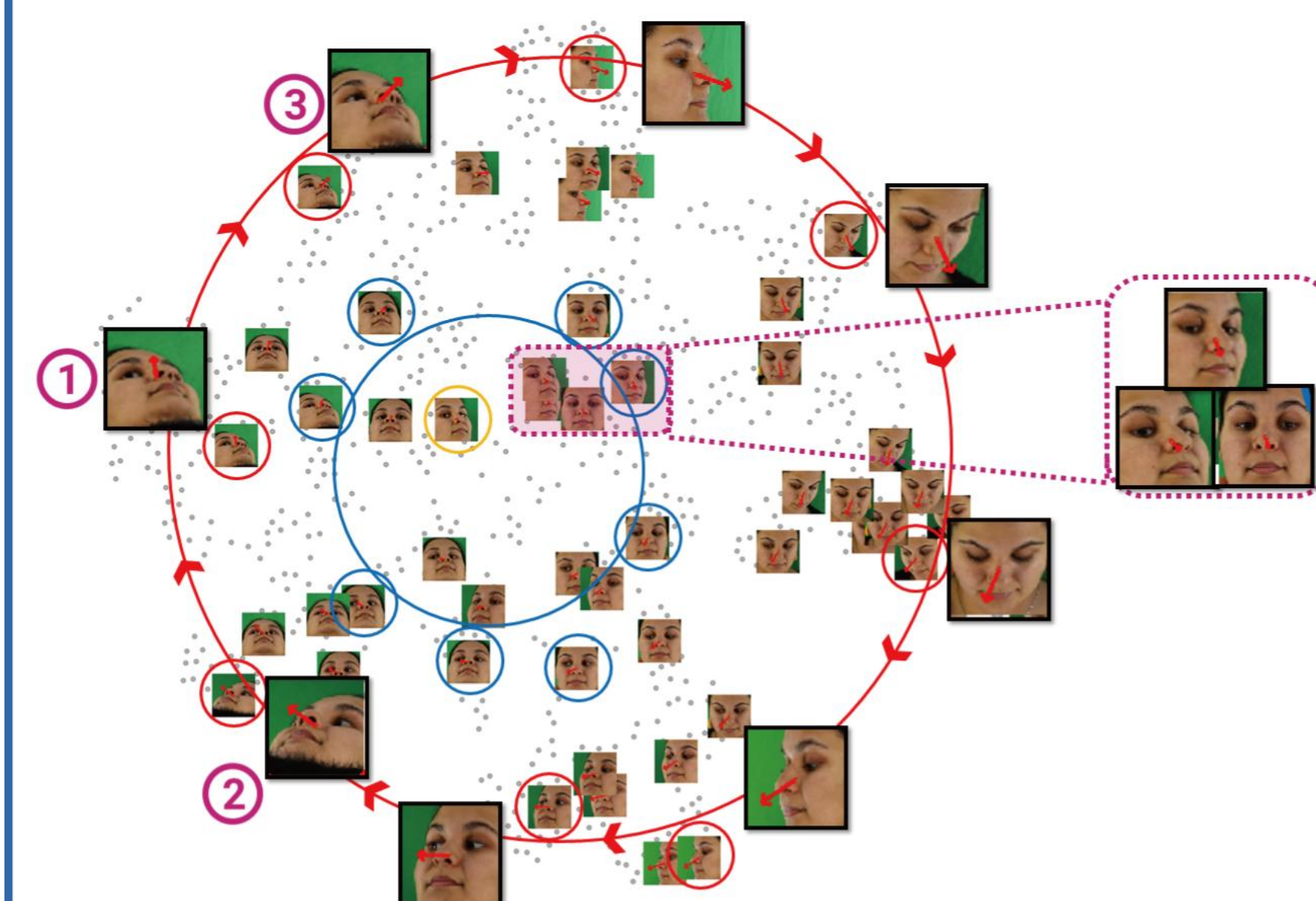
Ablation study

L_{gaze}	L_{align}^C	L_{align}^N	$\mathcal{D}_E \rightarrow \mathcal{D}_M$	$\mathcal{D}_E \rightarrow \mathcal{D}_D$	$\mathcal{D}_G \rightarrow \mathcal{D}_M$	$\mathcal{D}_G \rightarrow \mathcal{D}_D$
✓			8.07	8.78	7.94	8.73
✓	✓		7.14	7.95	7.26	8.11
✓	✓	✓	6.58	7.18	6.57	7.57

- ✓ SeeTN consistently improves both backbones.
- ✓ SeeTN also improves within-domain accuracy.

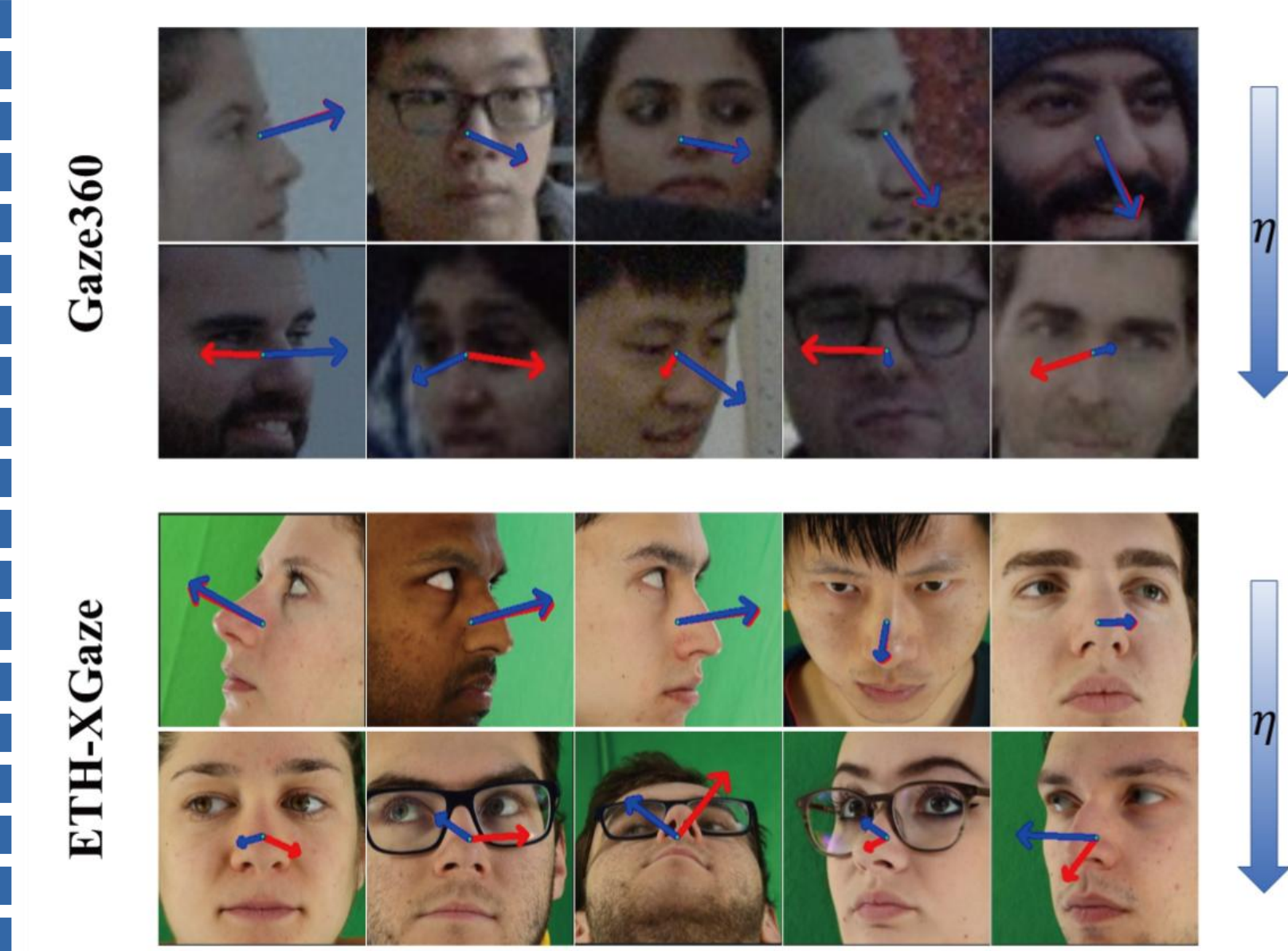
B) Qualitative Analysis

Feature visualization (t-SNE)



SeeTN learns **richer** gaze-semantic features.

Noisy sample detection (sorted by η)



Noisy samples receive **larger** η values.

4 Experimental Settings

